Experimental Probability

Name _____ Date _____ No calculator Version 1

KEY WORDS experiment · trial · frequency · relative frequency · experimental probability · theoretical probability

A Relative frequency ●○○

WORK IN YOUR BOOK

Remember: Experimental probability $= \frac{\text{number of times it happened}}{\text{number of trials}}$. This is also called the relative frequency.

1. Find the experimental probability as a fraction in its simplest form.

a) A coin is tossed 20 times and lands on heads 12 times. Find the probability of heads.

b) A dice is rolled 30 times and shows a six 6 times. Find the probability of a six.

c) A netball player scores from 15 of her 20 shots. Find the probability that she scores.

d) It rained on 9 of the 30 days in June. Find the probability of rain on a day in June.

e) A spinner is spun 40 times and lands on red 10 times. Find the probability of red.

2. Find the experimental probability as a decimal.

a) A drawing pin is dropped 100 times and lands point up 62 times. Find the probability of point up.

b) In a survey of 50 cars, 20 were white. Find the probability that a car is white.

c) A goalkeeper saves 3 of 10 shots. Find the probability of a save.

B Results in a table ●●○

WORK IN YOUR BOOK

3. A spinner with four colours is spun many times. The table shows the results.

Colour	Red	Blue	Green	Yellow
Frequency	12	18	6	4

a) How many times was the spinner spun?

b) Find the experimental probability of blue.

c) Which colour came up most often?

d) About how many times would you expect blue in 80 spins?

4. A bag holds counters of three colours. A counter is picked, its colour is written down and it is put back. The table shows the results.

Colour	Black	White	Grey
Frequency	25	15	10

a) How many times was the spinner spun?

b) Find the experimental probability of white.

c) Which colour came up most often?

d) About how many times would you expect white in 200 spins?

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C Using results to predict ●●● WORK IN YOUR BOOK

Remember: An experimental probability can be used to predict what will happen in more trials. If something happened in 0.3 of the trials, expect it in about 0.3 of any number of trials.

5. A bottle top is flipped 50 times. It lands the right way up 20 times.

a) Estimate the probability that it lands the right way up.

b) About how many times would it land the right way up in 200 flips?

6. Mia shoots 40 goals at netball practice and scores 28 of them.

a) Estimate the probability that she scores.

b) About how many goals would she score from 100 shots?

D Experiment and theory ●●● WORK IN YOUR BOOK

Remember: The theoretical probability is what should happen when every outcome is equally likely. The experimental probability is what did happen. They are usually close, but not exactly the same.

7. A fair coin is tossed 40 times. It lands on heads 23 times.

a) Find the experimental probability of heads.

b) Write down the theoretical probability of heads.

c) How many heads would you expect in 40 tosses?

d) The coin is fair. Explain why the number of heads is not exactly 20.

8. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	9	11	10	8	12	10

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

★ Extension FOR EARLY FINISHERS

9. Ana tosses a coin 10 times and gets 7 heads. Ben tosses the same coin 500 times and gets 262 heads. Find each experimental probability of heads as a decimal. Whose result tells you more about whether the coin is fair? Explain.

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Experimental Probability

Name _____ Date _____ No calculator Version 2

KEY WORDS experiment · trial · frequency · relative frequency · experimental probability · theoretical probability

A Relative frequency ●○○

WORK IN YOUR BOOK

Remember: Experimental probability $= \frac{\text{number of times it happened}}{\text{number of trials}}$. This is also called the relative frequency.

1. Find the experimental probability as a fraction in its simplest form.

a) A coin is tossed 20 times and lands on heads 8 times. Find the probability of heads.

b) A dice is rolled 30 times and shows a six 3 times. Find the probability of a six.

c) A netball player scores from 16 of her 20 shots. Find the probability that she scores.

d) It rained on 12 of the 30 days in June. Find the probability of rain on a day in June.

e) A spinner is spun 40 times and lands on red 30 times. Find the probability of red.

2. Find the experimental probability as a decimal.

a) A drawing pin is dropped 100 times and lands point up 58 times. Find the probability of point up.

b) In a survey of 50 cars, 15 were white. Find the probability that a car is white.

c) A goalkeeper saves 4 of 10 shots. Find the probability of a save.

B Results in a table ●●○

WORK IN YOUR BOOK

3. A spinner with four colours is spun many times. The table shows the results.

Colour	Red	Blue	Green	Yellow
Frequency	20	10	6	4

a) How many times was the spinner spun?

b) Find the experimental probability of red.

c) Which colour came up most often?

d) About how many times would you expect red in 80 spins?

4. A bag holds counters of three colours. A counter is picked, its colour is written down and it is put back. The table shows the results.

Colour	Black	White	Grey
Frequency	10	25	15

a) How many times was the spinner spun?

b) Find the experimental probability of grey.

c) Which colour came up most often?

d) About how many times would you expect grey in 200 spins?

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C Using results to predict ●●● WORK IN YOUR BOOK

Remember: An experimental probability can be used to predict what will happen in more trials. If something happened in 0.3 of the trials, expect it in about 0.3 of any number of trials.

5. A bottle top is flipped 50 times. It lands the right way up 15 times.

a) Estimate the probability that it lands the right way up.

b) About how many times would it land the right way up in 200 flips?

6. Mia shoots 40 goals at netball practice and scores 32 of them.

a) Estimate the probability that she scores.

b) About how many goals would she score from 100 shots?

D Experiment and theory ●●● WORK IN YOUR BOOK

Remember: The theoretical probability is what should happen when every outcome is equally likely. The experimental probability is what did happen. They are usually close, but not exactly the same.

7. A fair coin is tossed 60 times. It lands on heads 27 times.

a) Find the experimental probability of heads.

b) Write down the theoretical probability of heads.

c) How many heads would you expect in 60 tosses?

d) The coin is fair. Explain why the number of heads is not exactly 30.

8. A dice is rolled 60 times. The table shows the results.

Number	1	2	3	4	5	6
Frequency	11	9	10	12	8	10

a) Find the experimental probability of rolling a 6.

b) Write down the theoretical probability of rolling a 6 on a fair dice.

c) How many times would you expect each number in 60 rolls of a fair dice?

d) Do the results suggest that this dice is fair? Give a reason.

★ Extension FOR EARLY FINISHERS

9. Ana tosses a coin 10 times and gets 3 heads. Ben tosses the same coin 500 times and gets 241 heads. Find each experimental probability of heads as a decimal. Whose result tells you more about whether the coin is fair? Explain.

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